HOME SERVICE CONTACT

SNORT

INTRUSION DETECTION
SYSTEM



CONTENT

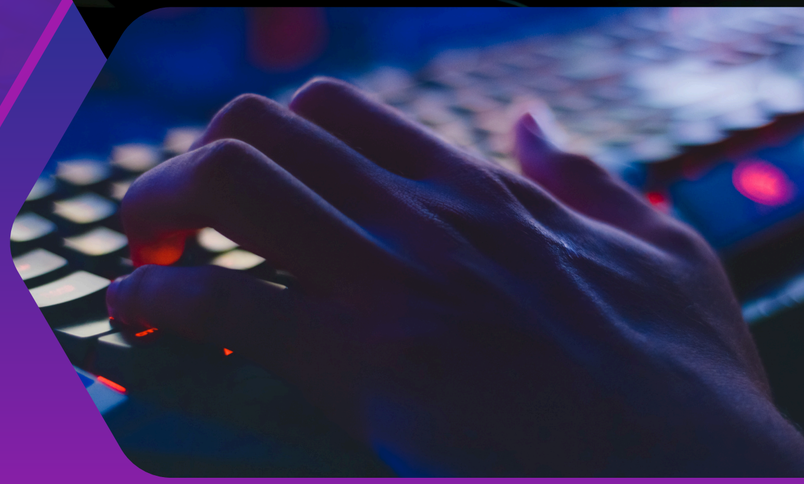
PART 1 : FUNDAMENTALS OF SNORT

- Introduction
- Rule
- Snort Architecture

PART 2: PRACTICAL

- Installation
- Configure Snort Rule
- Intrusion Detection with Snort



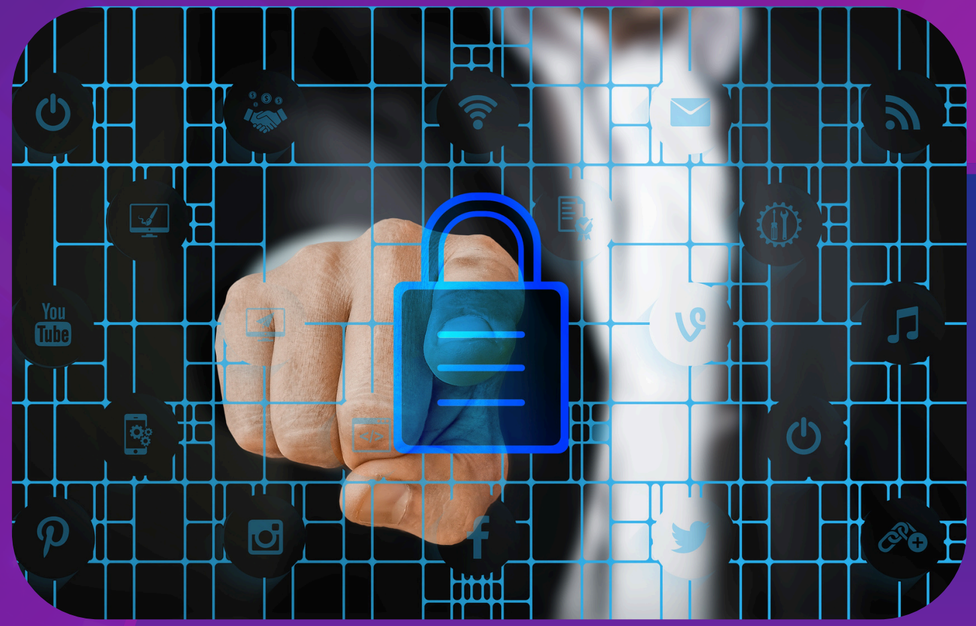


PART 1: FUNDAMENTALS OF SNORT

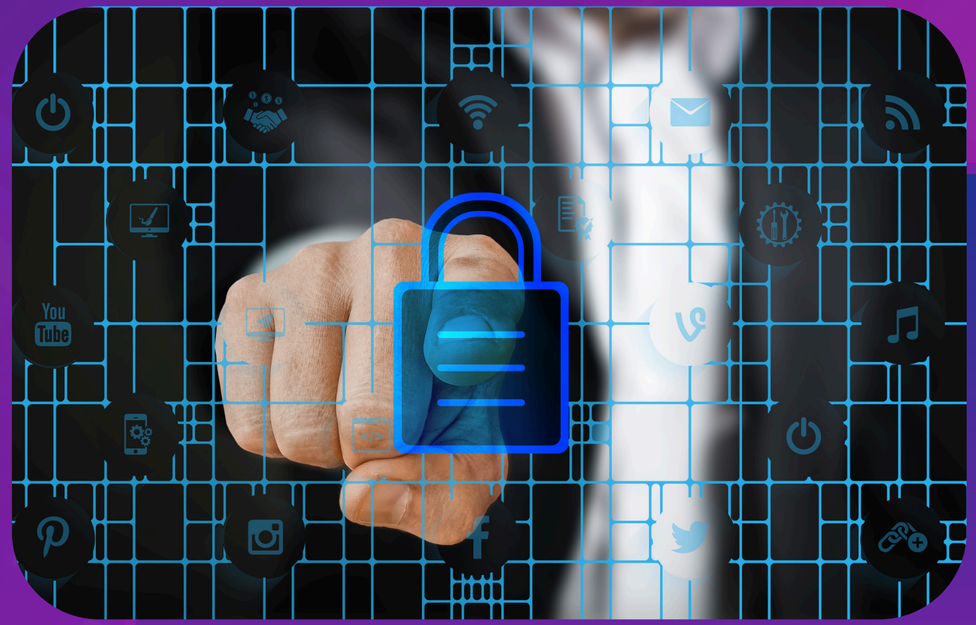
Introduction

- Snort is an open-source network intrusion detection and prevention system (IDS/IPS) that monitors network traffic and identifies potentially malicious activities on Internet Protocol (IP) networks.

- Originally developed by Martin Roesch in 1998 and is now maintained by Cisco's Talos Security Intelligence and Research Group.



- Snort detects malicious traffic or attacks by leveraging pattern matching.
- When active, Snort captures packets, reassembles them, analyzes them and determines what needs to be done to the packet based on predefined rules.
- Snort rules are very similar to a typical firewall rule, whereby, they are used to match network activity against specific patterns or signatures and consequently make a decision as to whether to send an alert or drop the traffic (in the case of IPS).



Snort Features

SIGNATURE-BASED DETECTION

- Uses predefined rules (signatures) to identify known attacks.

PACKET SNIFFER

- Captures and monitors network traffic in real time.

PACKET LOGGER

- Saves network traffic to files for later analysis.

FLEXIBLE RULE LANGUAGE

- Admins can write custom rules for specific threats or policies

PROTOCOL ANALYSIS

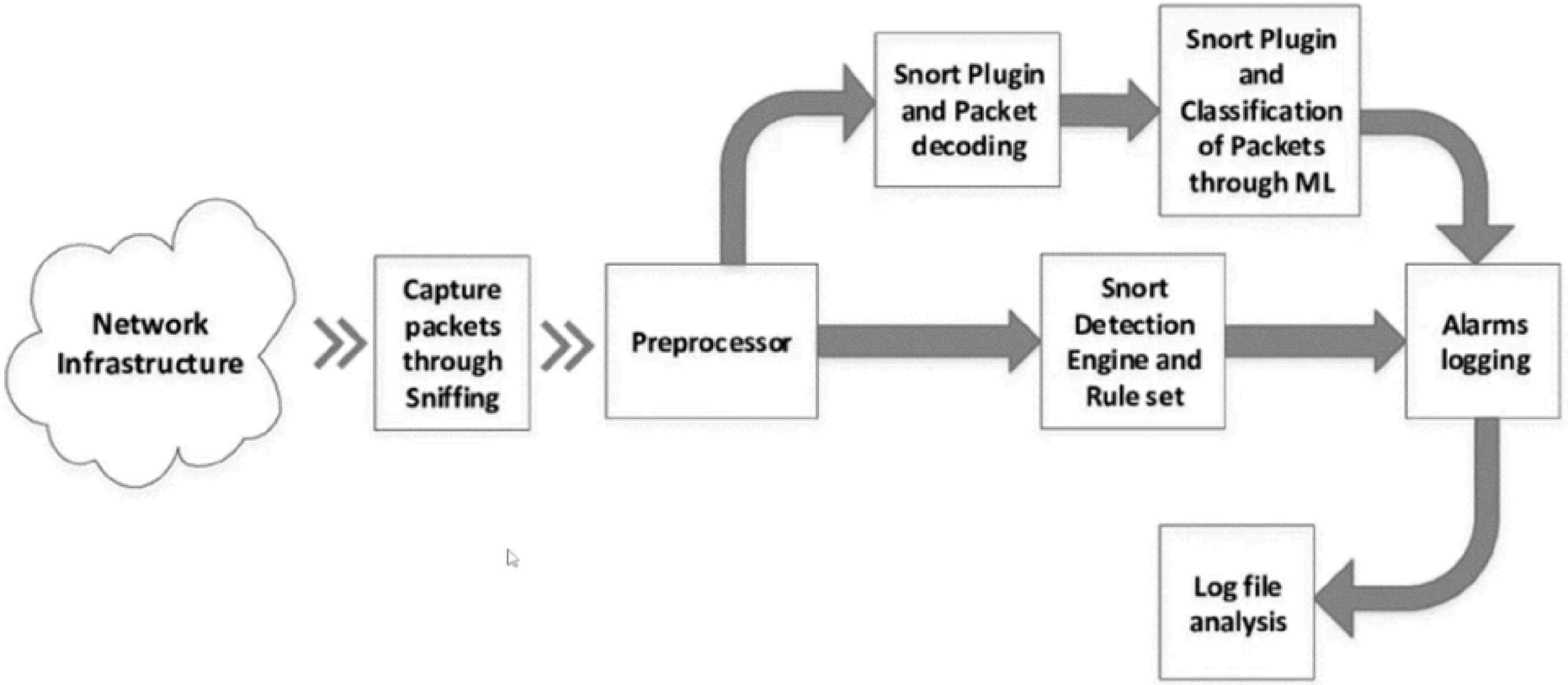
- Understands and analyzes network protocols(TCP, UDP, HTTP, etc.).

CROSS-PLATFORM SUPPORT

- Runs on both Linux and Windows

SNORT ARCHITECTURE

How Snort works



NETWORK INFRASTRUCTURE

- Snort sits on a network and monitors all traffic that flows through it.
- This traffic can be legitimate (normal use) or malicious (attacks, scans, exploits).

CAPTURE PACKETS THROUGH SNIFFING

- Snort works like a packet sniffer (similar to Wireshark).
- It captures raw packets from the network for inspection.

PREPROCESSOR

- Before analysis, the packets go through a preprocessor.
- This step normalizes the traffic (removes noise, reassembles fragmented packets, detects anomalies).

SNORT PLUGIN AND PACKET DECODING

- Plugins decode packet headers and payloads (IP, TCP, UDP, ICMP, etc.).
- This makes it easier for Snort to understand what each packet is carrying.

SNORT PLUGIN & MACHINE LEARNING CLASSIFICATION (OPTIONAL)

- Some advanced Snort deployments use machine learning to classify packets and detect new or unknown attacks.
- This complements rule-based detection by identifying patterns not explicitly written in rules.

ALARMS LOGGING

- If malicious or suspicious activity is detected, Snort logs an alert.
- Alerts can be sent to the console, stored in log files, or forwarded to a SIEM (e.g., Splunk, ELK stack).

LOG FILE ANALYSIS

- Security analysts review logs and alerts.
- Helps in incident response, forensic analysis, and tuning Snort rules

Snort IDS network placement

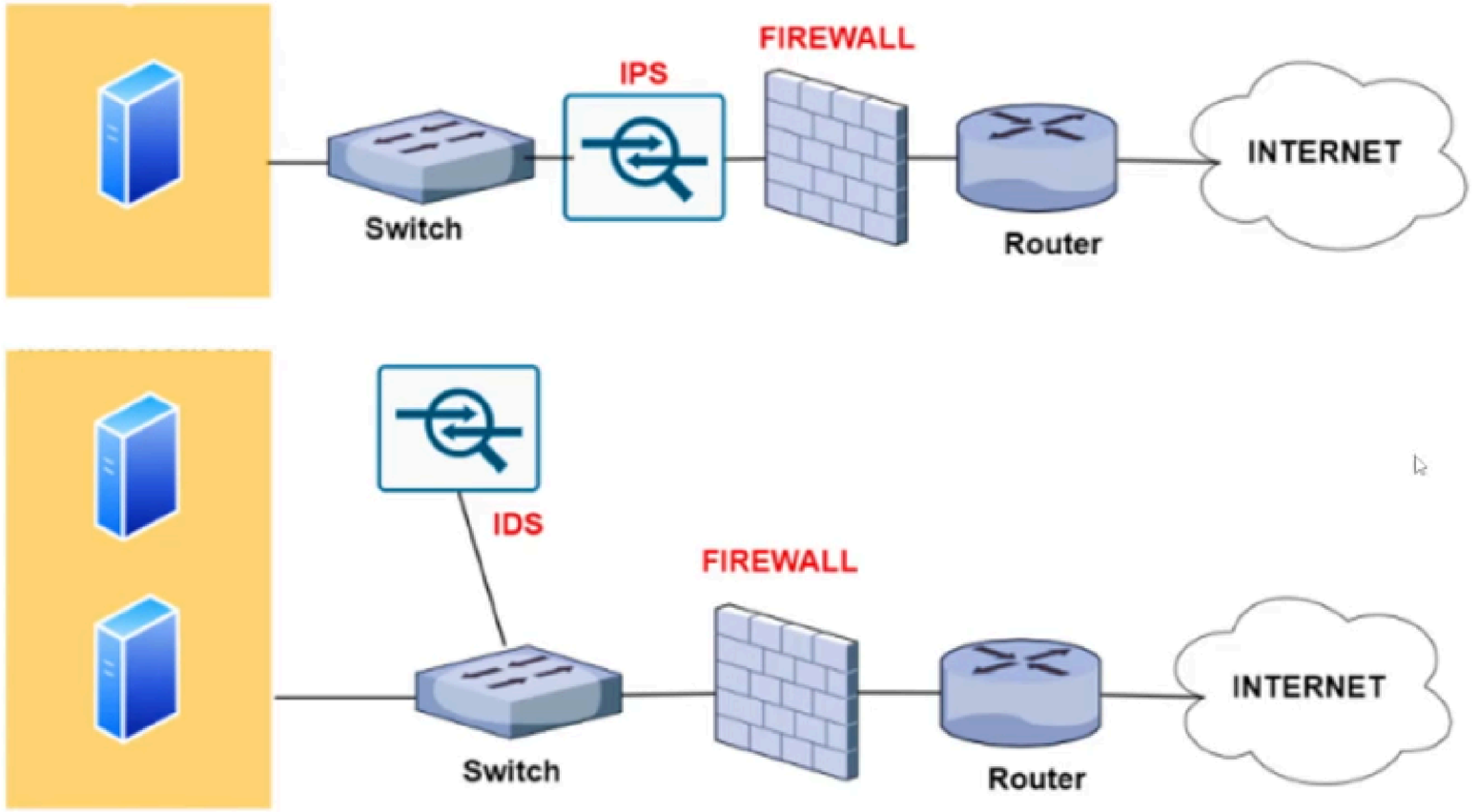


DIAGRAM EXPLAINATION

TOP PART - IPS

- Snort is acting as an IPS
- The IPS sits in-line (directly in the traffic path).
- It inspects packets in real time
- If it detects malicious traffic, it can block or drop it immediately before it reaches the firewall or router.

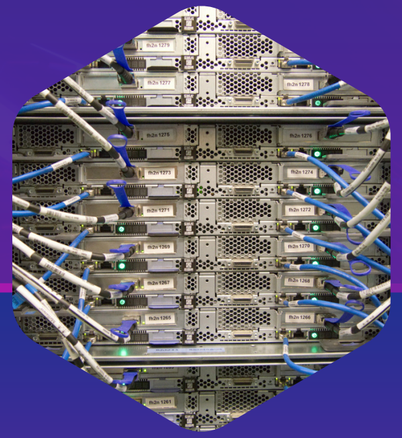
BOTTOM PART - IDS

- Snort is acting as a passive IDS.
- The IDS monitors copies of network traffic from the switch.
- It inspects packets in real time
- It detects and alerts about suspicious activities but cannot block traffic directly
- The firewall or admin must take action manually or through automation..



SNORT RULES

Snort Rules



COMMUNITY RULES

- Basic rules to get started
- Free rule-sets
- Created by Snort Community



REGISTERED RULES

- Free Rule-sets
- In order to use, Need to register for an account
- Created by Cisco Talos



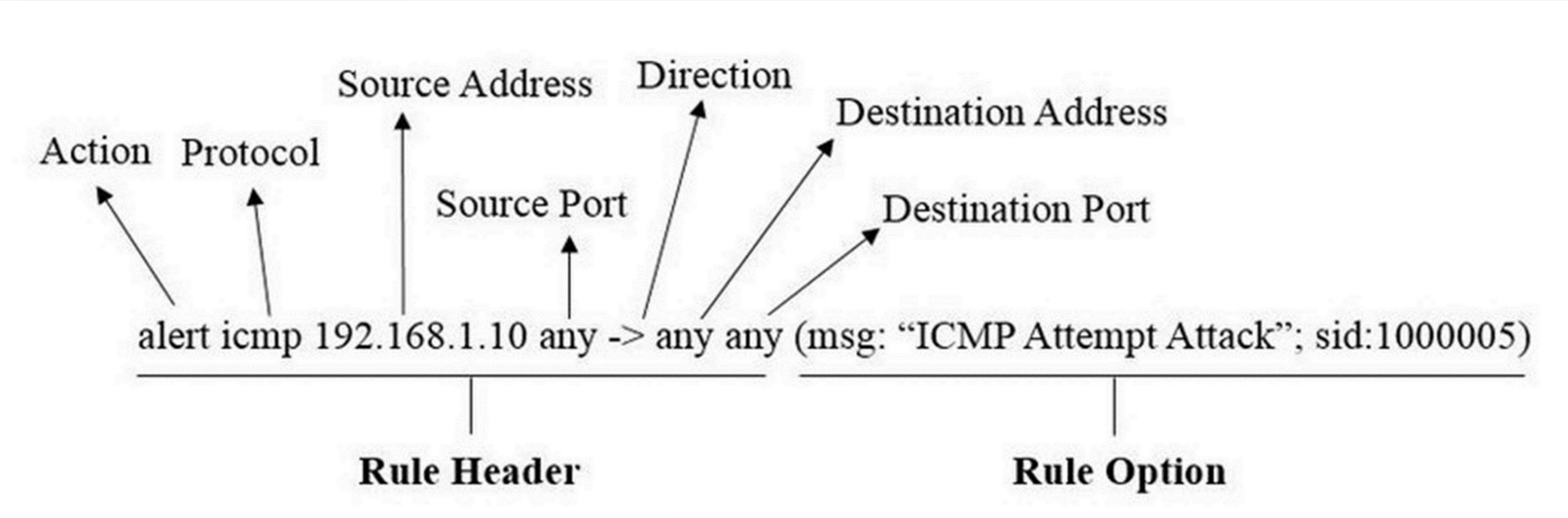
SUBSCRIPTION ONLY RULES

- Require active paid subscription in order to be accessed and used
- Get latest rule as soon as the rules released
- Used by Company and Security Team

PART 2: PRACTICALS

INSTALLATION

Snort Rule Syntax



*Refer to the **Snort Rule** Documentation for more information

Explaination

RULE HEADER

The rule header defines the basic matching criteria for the rule essentially the "when and where" to trigger the rule.

RULE OPTION

The rule option specifies detailed conditions for matching and what action to take essentially the "how" and "why" of the rule. It is enclosed in parentheses (...) and contains option keywords separated by semicolons.

SUMMARY

- Rule Header = The conditions for which packets are examined.
- Rule Option = The specific details and actions once those packets match.

Thank You

